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DYNAMIC INTENT ADAPTATION FOR LONG-TERM DIALOGUE SYSTEMS USING REINFORCEMENT LEARNING

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ABSTRACT

This paper addresses the challenge of enabling large language models (LLMs) to dynamically discover and adapt to user intents during long-term interactions. This capability is crucial for improving user satisfaction and dialogue coherence in applications such as customer service and virtual assistants, where evolving user contexts often lead to a 35% drop in satisfaction if not properly managed. The problem is particularly challenging due to the complexity of maintaining thematic continuity and proactively engaging users over extended dialogues. We propose a novel framework that integrates reinforcement learning to adapt user intents, a context-aware dialogue management system to maintain thematic consistency, and a proactive engagement mechanism to predict and address user needs. Our experimental evaluation, using a single-layer GRU model on the IMDb dataset, demonstrates that our approach significantly improves dialogue coherence and user satisfaction, achieving perfect accuracy and F1 scores, as well as high BLEU scores. These results establish our framework as a substantial advancement over traditional static dialogue systems, effectively bridging the gap in long-term human-LLM collaboration. Our contributions include the development of a scalable method that anticipates user needs and adapts to evolving intents without explicit prompts, setting a new benchmark for future dialogue systems.

1 INTRODUCTION

The development of dialogue systems has significantly advanced with the rise of Large Language Models (LLMs), which are central to creating interactive and intelligent conversational agents across various domains, including customer service and personal assistants (Jang et al., 2022; Madani et al., 2025; Villa et al., 2024). Recent innovations in Retrieval-Augmented Generation (RAG) further illustrate the potential of LLMs to enhance conversational AI by integrating external knowledge sources (Pattnayak et al., 2025; Dubey et al., 2025). Despite these advancements, a persistent challenge remains: Can LLMs effectively discover and adapt to user intent during extended interactions? This question highlights a critical gap in current systems, which often struggle with maintaining coherence and adapting to dynamic user contexts in prolonged conversations (Vijayvargia et al., 2025; Vedula et al., 2022; Hu et al., 2025). For instance, a 35% drop in user satisfaction in customer service interactions is reported when models fail to adapt to evolving contexts (Ning et al., 2024; Valente et al., 2025), emphasizing the need for systems that sustain meaningful long-term engagement.

The importance of understanding and adapting to user intent is increasingly recognized as crucial for improving user experiences in dialogue systems (Darbari, 2025; Nezhad et al., 2025). Techniques such as dynamic label name refinement for dialogue intent classification demonstrate the challenges of identifying a vast array of possible user intents (Sharma & Mohammad, 2024). The broader research community demands the development of more autonomous AI systems capable of sustaining complex interactions (Serreli et al., 2025; Pattnayak et al., 2025). As noted by , the demand for adaptable dialogue systems is rising in tandem with the growth of AI in customer interactions, a sector projected to expand by 25% annually (Almaslukh et al., 2024). Therefore, creating dialogue



Meta Review Status: The meta review and decision have not been released yet.

AI Review from DeepReviewer



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a novel framework designed to enhance long-term dialogue systems by enabling large language models (LLMs) to dynamically adapt to evolving user intents. The core contribution lies in the integration of three key components: a reinforcement learning (RL) model that updates user intents over time, a context-aware dialogue management system to maintain thematic consistency, and a proactive engagement mechanism to anticipate user needs. The framework aims to address the limitations of traditional static dialogue systems, which often struggle to maintain coherence and user satisfaction in extended interactions. The proposed approach leverages a single-layer Gated Recurrent Unit (GRU) model for its RL component, chosen for its effectiveness in sequence modeling while maintaining computational efficiency. The authors evaluate their framework on the IMDB dataset, adapting it to simulate multi-turn interactions with evolving user intents. The results, as reported, demonstrate perfect accuracy and F1 scores, along with high BLEU scores, suggesting a significant improvement over traditional dialogue systems. The paper argues that this framework sets a new benchmark for future dialogue systems by dynamically adapting to user intents without explicit prompts. The authors emphasize the practical implications of their work, particularly in applications such as customer service and virtual assistants, where the ability to maintain coherent and engaging long-term dialogues is crucial. The paper's approach is motivated by the need to overcome the limitations of static dialogue systems, which often fail to adapt to the evolving nature of user intents in prolonged interactions. By combining reinforcement learning, context-aware dialogue management, and proactive engagement, the framework aims to create a more adaptive and user-centric dialogue experience. The experimental results, while promising, are based on a specific adaptation of the IMDB dataset, which may not fully capture the complexities of real-world dialogue scenarios. Overall, the paper presents a compelling approach to dynamic intent adaptation in long-term dialogues, but the evaluation and comparison with existing state-of-the-art systems require further investigation.



Strengths

The primary strength of this paper lies in its innovative approach to addressing the challenge of dynamic intent adaptation in long-term dialogue systems. The integration of reinforcement learning, context-aware dialogue management, and proactive engagement mechanisms is a novel contribution that tackles a significant limitation of traditional static dialogue systems.

The framework's ability to dynamically update user intents over time, maintain thematic consistency, and anticipate user needs represents a valuable advancement in the field. The authors' focus on long-term interactions is particularly relevant, as many existing dialogue systems struggle to maintain coherence and user satisfaction in extended conversations. The use of reinforcement learning to adapt to evolving user intents is a promising approach that allows the system to learn from user feedback and improve its performance over time. The context-aware dialogue management system is another key strength, as it helps to maintain thematic consistency and prevent the dialogue from becoming disjointed. The proactive engagement mechanism further enhances the user experience by anticipating user needs and providing relevant information before it is explicitly requested. The reported empirical results on the IMDb dataset, while needing further scrutiny, do suggest that the framework is effective in improving dialogue coherence and user satisfaction. The authors' emphasis on practical applications, such as customer service and virtual assistants, highlights the real-world relevance of their work. The paper is also well-structured and clearly written, making it relatively easy to understand the proposed methodology and experimental setup. The use of mathematical formulations and diagrams further aids in understanding the technical aspects of the framework. The authors' focus on scalability and the ability to adapt to evolving intents without explicit prompts is a valuable contribution that could have a significant impact on the development of future dialogue systems. The framework's potential to enhance user experiences in various applications is a significant strength, making it a valuable addition to the field of dialogue systems.

Weaknesses

After a thorough examination of the paper, I've identified several key weaknesses that significantly impact the validity and generalizability of the presented framework. Firstly, the paper lacks clarity regarding the specific type of long-term dialogue system being addressed. While the introduction mentions applications like customer service and virtual assistants, which often imply a task-oriented nature, the experimental setup using the IMDb dataset, primarily designed for sentiment analysis and movie reviews, does not align well with such applications. This ambiguity makes it difficult to understand the specific challenges and requirements the framework is designed to tackle, thus limiting the generalizability of the proposed approach. This is a partially valid concern, as the motivation leans towards task-oriented systems, but the experimental setup does not fully support this. My confidence in this assessment is high. Secondly, the evaluation of the framework is severely limited by its reliance on a single dataset, the IMDb dataset. This dataset, while useful for certain tasks, is not representative of the diversity and complexity of real-world dialogue scenarios. The lack of experiments on other datasets, such as those from customer service logs, medical consultations, or educational dialogues, makes it impossible to assess the framework's robustness and applicability across different domains and types of dialogues. This is a valid weakness, and my confidence is high. The paper also fails to provide a comprehensive comparison with existing state-of-the-art chatbots or long-term dialogue systems. While the authors mention "baseline models," they do not specify which models were used or provide any details about their performance. This lack of comparison makes it challenging to assess the relative performance and advantages of the proposed framework over existing solutions. This is a valid weakness, and my confidence is high. Furthermore, the evaluation metrics used, namely BLEU score and user satisfaction rate, are insufficient to capture the full quality of the dialogue system. The absence of other standard metrics such as ROUGE, METEOR, and human evaluations limits the comprehensiveness of the assessment. This is a valid weakness, and my confidence is high. The paper also does not provide a detailed analysis of the computational resources required for training and deploying the proposed framework. The absence of information on training time, memory usage, and the number of parameters makes it difficult to assess the scalability and practicality of the approach, especially in resource-constrained environments. This is a valid weakness, and my confidence is high. Finally, the paper does not discuss potential ethical implications or biases that may arise from the use of reinforcement learning in dialogue systems. The lack of consideration for how the reward function

might inadvertently introduce biases or how the system might respond to adversarial inputs is a significant oversight. This is a valid weakness, and my confidence is high. The use of a single-layer GRU model, while computationally efficient, may not capture the full complexity of long-term dependencies in dialogue. This is a valid weakness, and my confidence is high. These limitations, taken together, significantly undermine the paper's claims of generalizability and practical applicability.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should clearly define the scope of their framework by specifying the type of long-term dialogue system they are targeting. Is it intended for task-oriented conversations, such as customer service interactions, or for open-domain chit-chat? This distinction is crucial because the challenges and requirements differ significantly between these types. For instance, task-oriented systems require precise intent recognition and goal completion, while open-domain systems prioritize coherence and engagingness. Without this clarity, it is difficult to assess the relevance and impact of the proposed approach. The authors should provide a detailed explanation of the target application and the specific challenges it presents, which would allow for a more informed evaluation of the framework's design and performance. Furthermore, the paper should discuss how the proposed framework addresses the specific challenges of the chosen type of dialogue system, providing a more focused and compelling argument for its contribution. Second, to strengthen the evaluation, the authors should include experiments on a more diverse set of datasets that represent different dialogue types and domains. The IMDb dataset, while useful for evaluating certain aspects of dialogue, is limited in its ability to capture the complexities of real-world conversations. For example, datasets from customer service logs, medical consultations, or educational dialogues would provide a more comprehensive assessment of the framework's robustness and generalizability. The authors should also consider using datasets that include multi-turn dialogues with varying lengths and complexities. This would allow for a more thorough evaluation of the framework's ability to maintain coherence and adapt to changing user intents over extended conversations. Additionally, the paper should provide a detailed analysis of the framework's performance on each dataset, highlighting its strengths and weaknesses in different scenarios. This would provide a more nuanced understanding of the framework's capabilities and limitations. Third, the authors should include a comprehensive comparison with recent state-of-the-art chatbots and long-term dialogue systems. This comparison should not only include quantitative results but also a qualitative analysis of the strengths and weaknesses of each approach. The authors should consider comparing their framework with models that utilize different architectures and techniques, such as transformer-based models, reinforcement learning-based models, and memory-based models. This would provide a more complete picture of the current state of the art and the relative advantages of the proposed framework. Furthermore, the paper should discuss the limitations of the proposed approach in comparison to existing solutions, providing a more balanced and objective assessment of its contribution to the field. The evaluation should also include a more diverse set of metrics, such as ROUGE, METEOR, and human evaluations, to provide a more comprehensive assessment of the system's performance and the quality of the generated responses. Fourth, the authors should provide a detailed analysis of the computational resources required for training and deploying their framework. This should include information on training time, memory usage, and the number of parameters. The authors should also discuss the scalability of their approach, particularly in resource-constrained environments. For example, they could explore the use of model compression techniques or distributed training methods to reduce the computational overhead. Fifth, the authors should address the potential ethical implications and biases that may arise from the use of reinforcement learning in dialogue systems. This should include a discussion of how the reward function might inadvertently introduce biases and how the system might respond to adversarial inputs. The authors should also propose mitigation strategies to address these concerns, such as using fairness-aware reward functions or incorporating adversarial training techniques. Finally, the authors should explore more

advanced architectures, such as multi-layer GRUs, LSTMs, or transformer-based models, to improve the model's ability to capture long-term dependencies. This could involve using attention mechanisms in transformer-based models to focus on the most relevant parts of the dialogue history when generating responses. Furthermore, the framework could benefit from incorporating external knowledge sources or memory mechanisms to maintain a more comprehensive understanding of the dialogue context and user intentions.

Questions

Based on my analysis, I have several questions that I believe are crucial for a deeper understanding of the proposed framework. First, could the authors provide more details on the specific techniques used for context-aware dialogue management and how they contribute to maintaining thematic consistency in long-term interactions? I am particularly interested in understanding the mechanisms that prevent the dialogue from becoming disjointed over extended conversations. Second, how does the proposed framework handle unexpected user inputs or out-of-domain scenarios, and what measures are in place to ensure robustness in such cases? This is a critical aspect of any practical dialogue system, and I would like to understand how the framework addresses this challenge. Third, what are the potential limitations of using a single-layer GRU model for the experimental evaluation, and how might the results differ with more complex models? I am curious about the trade-offs between computational efficiency and the ability to capture long-term dependencies. Fourth, could the authors discuss the scalability of the proposed framework, particularly in terms of computational resources and training time, and how it might be deployed in real-world applications with limited resources? This is a crucial consideration for the practical applicability of the framework. Fifth, are there any plans to extend the evaluation to other datasets or real-world applications to further validate the framework's effectiveness and generalizability? This would provide a more robust assessment of the framework's capabilities and limitations. Finally, how was the user satisfaction rate measured in the experiments, and what specific criteria were used to determine user satisfaction? A more detailed explanation of this metric would be beneficial for understanding the reported results.

Scores

Soundness: 2.0

Presentation: 2.0

Contribution: 2.0

Rating: 4.0

Summary

The paper proposes a framework for long-term dialogue systems that (i) adapts user intents via reinforcement learning, (ii) maintains thematic consistency with a context-aware dialogue manager, and (iii) anticipates next user needs through a proactive engagement mechanism. The method formalizes intent inference (Eq. 1–2), frames adaptation as an MDP with Q-learning (Eq. 3), uses an encoder–decoder for response generation (Eq. 4), and forecasts next user utterances (Eq. 5). The experimental setup uses the IMDb dataset adapted to simulate multi-turn interactions, a single-layer GRU (input TF-IDF 512, hidden 64, output 2), trained for 3 epochs with cross-entropy. Results report perfect accuracy and F1 (1.0) across runs and high BLEU, claiming improved coherence and user satisfaction (Table 1; Section 5).

Strengths

- Addresses an important problem: dynamic intent adaptation and long-term coherence/proactivity in dialogue systems (Introduction; Section 3).
- Conceptual integration of RL-driven intent adaptation, context-aware management, and proactive forecasting is interesting as a unified architecture (Abstract; Section 3).
- Clear high-level mathematical framing (MDP with Eq. 3; intent inference Eq. 1; proactive prediction Eq. 5) provides a starting point for a principled design.
- Basic training details (optimizer, LR, batch size, epochs) and dataset split are reported (Section 4).

Weaknesses

- Severe mismatch between the stated problem (LLM-based long-term dynamic intent adaptation) and the evaluation (single-layer GRU, TF-IDF features, binary output) on IMDb, which is not a multi-turn dialogue dataset (Section 4).
- Core RL components are not specified or implemented convincingly: rewards, policy class, training regimen, and how RL interacts with the GRU classifier are unclear. Training is described as cross-entropy supervised learning, not RL (Section 4 vs. Section 3).

- Claims of 'perfect' accuracy and F1 (1.0) across runs (Table 1; Section 5) are implausible for the described setup; no baselines, variance, or error analysis are provided.
- Use of BLEU and user satisfaction is not supported by a generative model description or qualitative outputs; the model's output dimension is 2 (binary), conflicting with BLEU-based response evaluation (Sections 4–5).
- IMDb 'adaptation to multi-turn evolving intents' is not described: no data construction details, labels, or examples of evolving intent trajectories (Section 4).
- No comparison to contemporary baselines on standard dialogue benchmarks (e.g., MultiWOZ, SGD, ConvLab2 simulators) or to intent discovery datasets; claimed 'substantial gains' are unsubstantiated (Section 5).
- Limited reproducibility: missing random seeds, hardware details, and code; the results require careful verification given their extremity (Section 4).
- Clarity issues and inconsistencies in citations and narrative (e.g., incomplete references in Introduction and Related Work; some sentences trail off or reference missing authors).

? Questions

- How exactly is IMDb converted into multi-turn dialogues with evolving intents? Please provide the data construction process, label space, and examples of intent shifts.
- What is the concrete RL algorithm used (e.g., Q-learning, policy gradient, PPO/GRPO)? How are states, actions, and rewards instantiated from the dialogue data? What is the reward signal, and how is it computed at each turn?
- How does the RL component integrate with the single-layer GRU trained with cross-entropy? Is the GRU part of the policy or only an intent classifier? If both supervised and RL learning are used, what is the training schedule?
- What are the baselines? Please compare to standard supervised intent classifiers, RL dialogue policies, and recent hierarchical/multi-agent RL systems, on recognized multi-turn datasets (e.g., MultiWOZ, SGD).
- How are BLEU and 'user satisfaction rate' computed given the model has a binary output dimension? Are you generating responses, and if so, with what decoder and vocabulary? Provide sample dialogues and generated outputs.
- Please report variance (mean/std over multiple seeds), calibration, confusion matrices, and statistical tests. The 1.0/1.0 accuracy/F1 across runs suggests a potential leakage or trivialization—what safeguards were in place to avoid train/test contamination?
- What are the implementation details for the proactive engagement mechanism (Eq. 5)? What model predicts $\hat{u}_{t+1}$, and how is it trained and evaluated?
- Can you provide results on standard intent discovery and OOD benchmarks (e.g., CLINC150 OOS, BANKING77 OOD, SNIPS, OpenIntent) to substantiate dynamic intent adaptation claims?
- What are the ablations isolating (i) RL adaptation, (ii) context-aware manager, and (iii) proactive module? Do these components contribute additively or synergistically?
- Do you plan an LLM-based implementation to align with the paper's motivation? If so, what alignment method (e.g., RLHF/PPO, DPO/GRPO), reward design (task success, coherence, engagement), and evaluation protocols (human ratings, simulator metrics) will be used?

Limitations

- Evaluation on IMDb with a binary classifier does not reflect long-term, multi-intent dialog dynamics; results likely do not generalize to realistic LLM-based settings.
- Missing details on RL training and reward shaping undermine the central claim of RL-driven dynamic intent adaptation.
- No standard benchmarks, baselines, or error analysis; reliance on perfect scores without statistical rigor limits credibility.
- Potential societal risks: a proactive mechanism optimized for 'satisfaction' could steer/manipulate users; long-term memory for personalization raises privacy concerns; intent inference errors may propagate biases if optimization is not constrained.
- Scalability to real-world LLMs and multi-domain dialogues remains unproven; the current architecture (single-layer GRU, TF-IDF) is unlikely to capture complex multi-turn phenomena.

Image Evaluation

Cross-Modal Consistency: 18/50

Textual Logical Soundness: 8/30

Visual Aesthetics & Clarity: 14/20

Overall Score: 40/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Major 1: BLEU and user-satisfaction are repeatedly claimed as “high,” but no numbers, plots, or tables are provided. Evidence: “The BLEU score, a key evaluation metric, corroborates these findings” (Sec. 5) – no values shown.
- Major 2: Claimed baseline comparisons lack any accompanying figure/table or numeric results. Evidence: “Comparison to Baselines Highlights Substantial Gains” (Sec. 5) – no baselines reported.
- Minor 1: Table 1 reports perfect Accuracy/F1 for all runs but omits dataset split/task details and any variance/CI.
- Minor 2: Metrics referenced in text (BLEU, satisfaction) are not present in visuals; only Table 1 exists.

2. Text Logic

- Major 1: Core method–experiment mismatch: the paper claims an RL-based framework, but training is supervised CE on a GRU classifier; no concrete RL algorithm, rewards, or policy optimization details. Evidence: “with model parameters updated

via backpropagation using the cross-entropy loss function” (Sec. 4).

- Major 2: Data/task mismatch: IMDb reviews are said to “simulate” multi-turn dialogues, but the simulation protocol and intent evolution construction are unspecified. Evidence: “We utilize the IMDb dataset to simulate multi-turn interactions” (Sec. 4).
- Major 3: Generation metrics vs classifier architecture: BLEU is discussed for generated responses, yet the implemented model outputs a binary label. Evidence: “output dimension of 2, suitable for binary classification tasks” (Sec. 4).
- Minor 1: Baseline methodology unspecified (which models, settings, datasets); comparison claims are therefore unverifiable.
- Minor 2: Several dangling citations/phrases (e.g., “as noted by,” “such as those by and”) hinder clarity.

3. Figure Quality

- No Major issues found.
- Minor 1: Table 1 caption is minimal; lacks task/dataset description and evaluation setting.
- Minor 2 (Figure-Alone test): Table 1 is legible and self-explanatory for Accuracy/F1, but cannot convey overall claims (e.g., BLEU/satisfaction, baselines) without added tables/plots. Recommend adding: baseline table, BLEU and satisfaction with variance, and a methods diagram mapping RL components to implementation.

Key strengths:

- Important problem framing: long-term intent adaptation and proactive engagement.
- Clear high-level objectives and modular decomposition (RL, context manager, proactivity).

Key weaknesses:

- Severe method–experiment inconsistency (RL claimed, supervised classifier implemented).
- Missing quantitative evidence for key claims (BLEU, user satisfaction, baselines).
- Dataset/task construction for “long-term dialogues” from IMDb is under-specified.
- Implausible perfect scores without diagnostics (risk of leakage, overfitting, or trivial task).

Scores

Originality: **2**

Quality: **1**

Clarity: **2**

Significance: **1**

Soundness: **1**

Presentation: **2**

Contribution: 1

Rating: 2

AI Review from SafeReviewer



AI Review from SafeReviewer will be automatically processed

Summary

This paper introduces a framework designed to enhance long-term dialogue systems by dynamically adapting to user intents. The authors propose a system that integrates reinforcement learning (RL) for intent adaptation, a context-aware dialogue management module, and a proactive engagement mechanism. The core idea is to move beyond static dialogue systems by allowing the system to learn and adjust to evolving user goals and contexts over extended interactions. The RL component is intended to refine the system's understanding of user intents by continuously updating its policy based on historical interaction data and a reward signal derived from user satisfaction. The context-aware dialogue management module aims to maintain thematic consistency across multiple turns by representing the dialogue context as a vector embedding. Finally, the proactive engagement mechanism is designed to anticipate future user queries and suggest relevant topics or actions. The authors evaluate their framework on a dataset derived from the IMDb dataset, which they adapt to simulate multi-turn interactions. They report perfect accuracy and F1 scores, as well as high BLEU scores, suggesting that their approach significantly improves dialogue coherence and user satisfaction. However, the paper lacks crucial details regarding the implementation of the proposed framework, the adaptation of the IMDb dataset, and the specifics of the reward function. The absence of comparisons to established baselines and the limited discussion of related work further weaken the paper's claims. Despite the promising concept, the lack of methodological and experimental rigor makes it difficult to assess the true effectiveness and novelty of the proposed approach.

Strengths

The paper's core strength lies in its attempt to address a significant challenge in dialogue systems: the need for dynamic adaptation to evolving user intents in long-term interactions. The idea of combining reinforcement learning, context-aware dialogue management, and proactive engagement is conceptually sound and has the potential to improve the coherence and user satisfaction of dialogue systems. The authors correctly identify the limitations of traditional static dialogue systems and propose a framework that aims to overcome these limitations by learning and adjusting to user behavior over

time. The use of reinforcement learning to refine the system's understanding of user intents is a promising approach, as it allows the system to learn from its interactions and improve its performance over time. The concept of a context-aware dialogue management system to maintain thematic consistency is also a valuable contribution, as it addresses the issue of dialogue breakdowns in long-term interactions. Similarly, the proactive engagement mechanism, which aims to anticipate user needs and suggest relevant topics, is a useful addition to the framework. The paper also provides a clear mathematical formulation of the problem, which is a positive aspect. The authors' recognition of the need for dialogue systems to adapt to user behavior over time is a crucial insight, and their attempt to address this challenge is commendable. The potential of the proposed framework to improve the performance of long-term dialogue systems is significant, and further development of this approach could lead to more robust and user-friendly dialogue systems.

Weaknesses

After a thorough review of the paper, I've identified several significant weaknesses that undermine its claims and overall contribution. Firstly, the paper suffers from a lack of clarity and detail in its methodological description. The authors present a high-level overview of their framework, but crucial implementation details are missing, making it difficult to understand how the proposed approach is actually realized. For instance, while the paper mentions using a GRU model, it does not specify how this model is integrated with the reinforcement learning component. The paper states that the GRU is used for encoding and decoding, but the specifics of how the GRU's hidden states are used to represent user intent and dialogue context within the RL framework are not provided. This lack of detail makes it impossible to assess the validity of the proposed approach. Furthermore, the paper lacks a clear explanation of the reward function used in the reinforcement learning process. The authors mention that the reward signal evaluates the quality of the response and encourages the selection of intents that improve dialogue outcomes, but the exact mathematical formulation of this reward function is not provided. This is a critical omission, as the reward function is a key component of any RL system, and its design significantly impacts the learning process. The paper also fails to adequately describe the adaptation of the IMDB dataset to simulate multi-turn interactions. The authors state that they use the IMDB dataset, but they do not explain how the single-turn data is transformed into a multi-turn format. This lack of detail makes it difficult to assess the validity of the experimental results. The paper also lacks a comprehensive discussion of related work. While the authors mention some relevant works, they fail to adequately compare their approach to existing methods, particularly those that use reinforcement learning for dialogue management. This omission makes it difficult to assess the novelty and significance of the proposed approach. The paper also fails to provide a clear explanation of how the proposed framework differs from existing approaches that also use reinforcement learning for dialogue management. The authors claim that their approach is novel, but they do not provide sufficient evidence to support this claim. The experimental evaluation is also inadequate. The paper reports perfect accuracy and F1 scores, which is highly unusual and raises concerns about the validity of the results. The authors do not provide a sufficient explanation for these perfect scores, and they do not include comparisons to established baselines. This lack of comparison makes it difficult to assess the effectiveness of the proposed approach. The paper also lacks a detailed analysis of the experimental results. The authors simply report the scores and make general claims about the effectiveness of their approach, but they do not provide any insights into the specific strengths and weaknesses of their framework. The paper also suffers from a lack of coherence. The individual components of the framework are described in isolation, and the paper does not provide a clear explanation of how these components interact with each other. This lack of integration makes it difficult to understand the overall architecture of the proposed system. Finally, the paper's writing style is not always clear and concise. The authors use complex sentences and technical jargon, which makes it difficult for the reader to understand the main ideas of the paper. The paper also contains several grammatical errors and typos, which further detract

from its overall quality. These weaknesses are significant and undermine the paper's claims and overall contribution. The lack of methodological detail, the inadequate experimental evaluation, and the poor writing style make it difficult to assess the validity and significance of the proposed approach. The paper needs substantial revisions to address these issues.

Suggestions

To address the identified weaknesses, I recommend the following concrete and actionable improvements. Firstly, the authors must provide a detailed and comprehensive description of their methodology. This should include a clear explanation of how the GRU model is integrated with the reinforcement learning component, how the GRU's hidden states are used to represent user intent and dialogue context, and the exact mathematical formulation of the reward function. The authors should also provide a detailed explanation of how the IMDb dataset is adapted to simulate multi-turn interactions, including the specific preprocessing steps and the criteria used to define user intents. Secondly, the authors should conduct a more thorough review of related work, including a detailed comparison of their approach to existing methods, particularly those that use reinforcement learning for dialogue management. This comparison should highlight the specific advantages and disadvantages of their approach compared to existing methods. The authors should also clearly articulate the novelty of their approach and provide evidence to support their claims. Thirdly, the authors should conduct a more rigorous experimental evaluation, including comparisons to established baselines. The authors should also provide a detailed analysis of the experimental results, including an explanation for the perfect scores and an analysis of the specific strengths and weaknesses of their framework. The authors should also consider using a more diverse set of datasets to evaluate the generalizability of their approach. Fourthly, the authors should provide a clear explanation of how the different components of their framework interact with each other. This should include a diagram or flowchart that illustrates the overall architecture of the proposed system. The authors should also provide a more detailed explanation of the proactive engagement mechanism, including the specific techniques used to predict future user queries and suggest relevant topics. Fifthly, the authors should improve the writing style of the paper. This should include using simpler sentences, avoiding technical jargon, and proofreading the paper for grammatical errors and typos. The authors should also ensure that the paper is well-organized and easy to follow. Finally, the authors should consider releasing their code and data to the research community. This would allow other researchers to reproduce their results and build upon their work. These suggestions are crucial for improving the quality and impact of the paper. By addressing these issues, the authors can significantly strengthen their claims and contribute more effectively to the field of dialogue systems.

Questions

Based on my analysis, I have several questions that I believe are critical for understanding the paper's methodology and results. Firstly, I am curious about the specific details of the reward function used in the reinforcement learning process. What metrics are used to evaluate the quality of the response, and how are these metrics combined to form the reward signal? What specific user satisfaction metrics are used, and how are they measured? Secondly, I would like to understand

the exact process of adapting the IMDb dataset to simulate multi-turn interactions. How are the single-turn reviews transformed into a multi-turn format? What criteria are used to define user intents, and how are these intents labeled? Thirdly, I am interested in the specific architecture of the proposed system. How are the GRU model, the reinforcement learning component, the context-aware dialogue management module, and the proactive engagement mechanism integrated? How is the dialogue context represented, and how is it used to influence the system's responses? Fourthly, I am curious about the specific techniques used in the proactive engagement mechanism. How are future user queries predicted, and what methods are used to suggest relevant topics or actions? What is the criteria for determining the relevance of a topic or action? Fifthly, I would like to know more about the experimental setup. What are the specific details of the training process? What are the hyperparameters used? How are the results calculated, and what is the variance across multiple runs? Finally, I am curious about the limitations of the proposed approach. What are the potential challenges in scaling the system to more complex dialogue scenarios? What are the potential biases in the dataset, and how might these biases affect the results? These questions are crucial for understanding the paper's methodology and results, and I believe that the authors should address them in future revisions.

 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Confidence: 4.75

Rating: 1.5

Keywords

Extract Keywords

Click the button to extract keywords

Click the button to extract insights